

Endpoint Separation and the Polynomial–Analytic Interface in the Belgian Chocolate Problem

Supplementary proof and adversarial interface audit

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Abstract

The open-disc analytic feasibility predicate includes its critical parameter, whereas the original strictly Hurwitz-stable polynomial predicate does not. We prove the precise interface without altering the frozen Core Algorithm. The two new ingredients at the interface are a finite polynomial realization with strict parameter slack, and an explicit upward perturbation of the original parameter, valid even when the defining identity has leading-term cancellation. Combined with the accepted Core results, these give the original admissible set exactly as $(0, (1 - Q)/(1 + Q))$. The endpoint is excluded by an independent polynomial construction, not by a false same-parameter equivalence or by the definition of an infimum.

1 Original problem, notation, and scope

A nonzero polynomial in $\mathbb{R}[s]$ is *Hurwitz stable* if every complex zero has strictly negative real part. Nonzero constants are stable. For $\delta > 0$, put

$$a_\delta(s) = s^2 - 2\delta s + 1, \quad b(s) = s^2 - 1.$$

The original admissible set is

$$\mathcal{A}_{\text{poly}} = \left\{ \delta > 0 : \begin{array}{l} x, y, p \in \mathbb{R}[s] \setminus \{0\} \text{ are stable, } \deg x \geq \deg y, \\ p = a_\delta x + by \end{array} \right\}. \quad (1)$$

Here p denotes the polynomial called $z(s)$ in the original problem; z below is a complex variable. No sign or monic normalization of the leading coefficients is imposed. In particular, $\deg p$ need not equal $\deg x + 2$.

Let $I = (0, 1)$, $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$, and

$$\delta(q) = \frac{1 - q}{1 + q}, \quad q(\delta) = \frac{1 - \delta}{1 + \delta} \quad (0 < \delta < 1). \quad (2)$$

These are mutually inverse, strictly decreasing bijections of I . For $0 < t < 1$, $\mathcal{F}_t$ consists of holomorphic functions $f : \mathbb{D} \rightarrow \mathbb{C}$ such that

- $f(\bar{z}) = \overline{f(z)}$;
- the only zero of f is 0, and it is simple;
- the only solutions of $f(z) = 1$ are $it, -it$, both simple.

For $q \in I$, define the distinct predicates

$$P(q) : \delta(q) \in \mathcal{A}_{\text{poly}}, \quad A(q) : \mathcal{F}_{\sqrt{q}} \neq \emptyset,$$

and their parameter sets $\mathcal{S}_P = \{q \in I : P(q)\}$ and $\mathcal{S}_A = \{q \in I : A(q)\}$.

This supplement proves the interface from these definitions. It does not reopen or modify the finite systems, coefficient bounds, compactness proof, certificate searches, or subdivision program of the Core Algorithm.

2 Exact frozen Core interface and main result

The following are accepted inputs, not conclusions reproved in this note. They refer to the frozen source identified in Appendix A.

C1: analytic realization. For $0 < q \leq 9/16$,

$$A(q) \iff H_N(q) \text{ is feasible for every integer } N \geq 0. \quad (3)$$

The definitions and proofs of finite semialgebraicity, effective uniform coefficient bounds, and the infinite-layer realization remain frozen.

C2: critical value. The Core’s independently specified certified interval program determines a unique real Q , with

$$0 < Q < \frac{1}{2}, \quad Q = \inf \mathcal{S}_A, \quad A(Q). \quad (4)$$

Its identification with the analytic infimum and its endpoint analytic object are accepted here.

C3: finite certificates and total precision algorithm. For rational $0 < q \leq 1/2$, a finite infeasible layer exists exactly when $q < Q$, and the Core’s finite rational positive certificate exists exactly when $q > Q$. The fixed two-test-point subdivision program produces rational strict brackets $L_j < Q < R_j$ with

$$0 \leq L_j < R_j \leq \frac{1}{2}, \quad R_j - L_j \leq \frac{1}{2}(2/3)^j,$$

and each prescribed stage terminates. These statements and the program are unchanged.

C4: no finite stabilization. The frozen non-stabilization result for a fixed H_N is retained. No finite layer is assumed to be a complete threshold test.

No statement about H_N outside its frozen parameter range is used.

Theorem 2.1 (Repaired interface and combined original conclusion). *Under the frozen Core inputs above, let $\Delta = (1 - Q)/(1 + Q)$. Then*

$$A(q) \iff Q \leq q < 1, \quad q \in I, \quad (5)$$

$$P(q) \iff Q < q < 1, \quad q \in I, \quad (6)$$

$$\mathcal{A}_{\text{poly}} = (0, \Delta), \quad \frac{1}{3} < \Delta < 1. \quad (7)$$

In particular $A(Q)$ holds and $P(Q)$ fails. Moreover,

$$P(q) \iff \exists r \in (0, q) \ A(r), \quad q \in I, \quad (8)$$

and $\mathcal{S}_A = \overline{\mathcal{S}_P}^I$, where the closure is relative to I . For every rational $q \in (Q, 1)$ there is a finite rational polynomial witness of (1) at $\delta(q)$.

The proof is organized as follows. The analytic threshold information used in the interface arguments is exactly (4).

$$\begin{array}{c}
P(q) \implies A(q) \quad (\text{Cayley identity}) \\
A(r), r < q \implies \text{closed-disc polynomial factors} \implies P(q) \\
P(q) \implies P(q') \text{ for some } q' < q \implies q > Q. \\
\hline
\text{Frozen exact computation of } Q \quad + \quad \text{these interface implications} \implies (7).
\end{array}$$

3 Original polynomials imply open-disc analytic feasibility

Lemma 3.1. *No $\delta \geq 1$ belongs to $\mathcal{A}_{\text{poly}}$.*

Proof. At $\delta = 1$, evaluating the identity at $s = 1$ gives $p(1) = 0$, contrary to stability. If $\delta > 1$, the roots $r_{\pm} = \delta \pm \sqrt{\delta^2 - 1}$ of a_{δ} satisfy $0 < r_- < 1 < r_+$. Thus

$$\frac{p(r_-)}{y(r_-)} = r_-^2 - 1 < 0, \quad \frac{p(r_+)}{y(r_+)} = r_+^2 - 1 > 0.$$

But p/y is a continuous nonzero real function on $(0, \infty)$, since both polynomials are stable. It cannot change sign. $\square$

Lemma 3.2 (Unconditional forward implication). *For every $q \in I$, $P(q)$ implies $A(q)$, regardless of degree cancellation in p .*

Proof. Take a witness at $\delta = \delta(q)$ and write

$$w = \frac{s-1}{s+1}, \quad s = \frac{1+w}{1-w}.$$

The second map is a biholomorphism from $\mathbb{D}$ onto the open right half-plane. Direct calculation gives

$$(s+1)^2 w = b(s), \quad (s+1)^2 (w^2 + q) = (1+q)a_{\delta}(s). \quad (9)$$

Define

$$f(w) = \frac{b(s)y(s)}{p(s)}, \quad 1 - f(w) = \frac{a_{\delta}(s)x(s)}{p(s)}. \quad (10)$$

Stability makes these holomorphic on $\mathbb{D}$. Real coefficients give real symmetry. In the right half-plane the only zero of b is the simple zero $s = 1$; neither y nor p has a zero there. Hence f has just one zero, simple and at $w = 0$. The roots of a_{δ} are $\delta \pm i\sqrt{1 - \delta^2}$, distinct and in the right half-plane. By (9), they correspond to $w = \pm i\sqrt{q}$. The factors x/p are nonvanishing there, and the change of variable has nonzero derivative. Thus these and only these are the simple 1-points.

The point $w = 1$, corresponding to infinity, is not in $\mathbb{D}$. No assertion about regularity there or about the degree of p was used. $\square$

4 Strict slack gives finite, exactly compatible polynomial factors

Lemma 4.1 (Scaling and strict margin). *If $0 < r < q < 1$ and $A(r)$, there are real-symmetric holomorphic, nonvanishing functions u, v on a disc of radius $R > 1$ satisfying*

$$zu(z) + (z^2 + q)v(z) = 1. \quad (11)$$

Proof. Take $f_0 \in \mathcal{F}_{\sqrt{r}}$, let $c = \sqrt{r/q} < 1$, and put $F(z) = f_0(cz)$. On $|z| < R = 1/c$, its only zero is the simple zero 0 and its only 1-points are the simple points $\pm i\sqrt{q}$. These points belong to that disc because $\sqrt{r} < 1$. The quotients

$$u(z) = \frac{F(z)}{z}, \quad v(z) = \frac{1 - F(z)}{z^2 + q}$$

have removable singularities at the indicated points. Simplicity and the absence of further zeros show that both quotients are nonvanishing throughout this disc. They have real Taylor coefficients and satisfy (11). In particular $v(0) = 1/q$. $\square$

Lemma 4.2 (Exact finite approximation). *Under the hypotheses of Lemma 4.1, there are nonzero real polynomials u_*, v_* , both zero-free on $\overline{\mathbb{D}}$, such that*

$$zu_*(z) + (z^2 + q)v_*(z) = 1. \quad (12)$$

If q is rational, they can both be chosen in $\mathbb{Q}[z]$.

Proof. The compactness of $\overline{\mathbb{D}}$ gives strictly positive numbers

$$m_u = \min_{\overline{\mathbb{D}}} |u|, \quad m_v = \min_{\overline{\mathbb{D}}} |v|.$$

Choose $0 < \eta < \frac{1}{2} \min\{m_v, m_u/(1+q)\}$. Since v is holomorphic on a larger disc, its Taylor polynomials converge uniformly on $\overline{\mathbb{D}}$. Choose one with error less than $\eta/2$. For real q it suffices to use this polynomial. For rational q , retain the constant coefficient $1/q$ exactly and approximate the finitely many other real coefficients by rationals so that the sum of their absolute errors is less than $\eta/2$. This is possible by density of $\mathbb{Q}$ in $\mathbb{R}$. Denote the resulting real or rational polynomial by v_* . Then

$$v_*(0) = 1/q, \quad \|v_* - v\|_{\overline{\mathbb{D}}} < \eta.$$

Define, rather than approximate independently,

$$u_*(z) = \frac{1 - (z^2 + q)v_*(z)}{z}.$$

The numerator has value zero at $z = 0$, so this is a polynomial and (12) is an exact identity. On $|z| = 1$,

$$|u_*(z) - u(z)| \leq (1+q) \|v_* - v\|_{\overline{\mathbb{D}}}.$$

The difference is holomorphic across 0 and on a neighborhood of the closed disc. The maximum modulus theorem therefore gives the same bound throughout $\overline{\mathbb{D}}$. Consequently

$$|v_*| > m_v - \eta > 0, \quad |u_*| > m_u - (1+q)\eta > 0 \quad \text{on } \overline{\mathbb{D}}.$$

All coefficients are real; for rational q they are rational. The exact identity also enforces the specified interpolation and symmetry, without any separate perturbation of the interpolation points. $\square$

Remark 4.3 (Existence versus effective enumeration). The margins in this proof depend on the particular r, q, f_0 and need not be uniform as $q \downarrow Q$. They are used to prove existence, not as uncomputable inputs to the frozen algorithm. For a given rational q , one may enumerate rational v_* with constant term $1/q$, compute u_* exactly, and decide closed-disc nonvanishing by real algebra. Explicitly, for each rational polynomial h , test the nonexistence of real a, b such that

$$a^2 + b^2 \leq 1, \quad \operatorname{Re} h(a + ib) = \operatorname{Im} h(a + ib) = 0.$$

These are finite rational polynomial conditions. Quantifier elimination for real closed fields is a terminating decision procedure for such conditions. The existence proof ensures that an exhaustive enumeration eventually finds a certificate when $q > Q$; it does not require computing the coefficients of f_0 or its margins. This observation verifies the interface and is not a modification of the frozen searches.

5 Finite factors give original stable polynomials

Lemma 5.1 (Exact Cayley realization and actual degrees). *Let $q \in I$ and let u_*, v_* satisfy Lemma 4.2. Set $n = \max\{\deg u_*, \deg v_*\}$ and $w = (s-1)/(s+1)$. Then*

$$\begin{aligned} x(s) &= (1+q)(s+1)^n v_*(w), \\ y(s) &= (s+1)^n u_*(w), \\ p(s) &= (s+1)^{n+2} \end{aligned} \tag{13}$$

are nonzero real stable polynomials satisfying $p = a_{\delta(q)}x + by$ and $\deg x = \deg y = n$. If q, u_, v_* are rational, all three have rational coefficients.*

Proof. For a nonzero polynomial h of degree $d \leq n$, factor $h(w) = h_d \prod_{j=1}^d (w - \omega_j)$. Then

$$(s+1)^n h\left(\frac{s-1}{s+1}\right) = h_d (s+1)^{n-d} \prod_{j=1}^d ((1-\omega_j)s - (1+\omega_j)). \tag{14}$$

Empty products cover constants. Since h is zero-free on $\overline{\mathbb{D}}$, every $|\omega_j| > 1$; in particular $\omega_j \neq 1$. The transformed roots satisfy

$$\operatorname{Re} \frac{1+\omega_j}{1-\omega_j} = \frac{1-|\omega_j|^2}{|1-\omega_j|^2} < 0.$$

The additional roots at -1 are also strictly stable. The coefficient of s^n is $h(1) \neq 0$, so the actual degree is exactly n , even if $d < n$. This proves the stability and degree claims for x, y ; those for p are immediate.

Using (9) and (12) gives

$$a_{\delta(q)}x + by = (s+1)^{n+2}((w^2 + q)v_*(w) + wu_*(w)) = (s+1)^{n+2}.$$

Initially this calculation is valid for $s \neq -1$; both sides are polynomials, so it is a polynomial identity everywhere. The displayed formulas prove real or rational coefficients without rounding the parameter. Every polynomial has finitely many strictly left-half-plane roots, hence a positive strict stability margin. $\square$

Lemmas 4.1–5.1 prove the general implication

$$0 < r < q < 1, A(r) \implies P(q). \tag{15}$$

By (4), take $r = Q$ for any $q > Q$. Thus off-endpoint realization is proved for every real $q \in (Q, 1)$, including parameters outside the range on which H_N is defined.

6 Polynomial perturbation without a degree-cancellation loop-hole

6.1 A quantitative fixed-degree stability neighborhood

Lemma 6.1. *For each nonzero stable polynomial h of actual degree d , there is $\tau_h > 0$ such that every real polynomial g of degree at most d whose coefficients differ from those of h by less than τ_h has degree exactly d and is stable.*

Proof. If $d = 0$, take $\tau_h = |h(0)|/2$. Otherwise choose mutually disjoint closed circular discs centered at the distinct roots of h , entirely in the open left half-plane, with boundaries containing no root. Let Γ be the union of their boundaries and set

$$\mu = \min_{s \in \Gamma} |h(s)| > 0, \quad B = \sum_{j=0}^d \max_{s \in \Gamma} |s|^j > 0, \quad \tau_h = \min \left\{ \frac{|h_d|}{2}, \frac{\mu}{2B} \right\}.$$

The coefficient condition preserves the nonzero leading coefficient and gives $|g - h| < |h|$ on every circle. Rouché's theorem states that two holomorphic functions with this strict boundary inequality have the same number of zeros, counted with multiplicity, inside the circle. It applies here since polynomials are holomorphic on and inside each circle. The enclosed multiplicities for h total d ; the same is true for g . Since $\deg g = d$, all its zeros lie in these left-half-plane discs. This also covers repeated roots. $\square$

6.2 The explicit perturbation

Theorem 6.2 (Strict upward perturbation in the original parameter). *If $\delta \in \mathcal{A}_{\text{poly}}$, there exists $\delta' \in \mathcal{A}_{\text{poly}}$ with $\delta < \delta' < 1$. This holds for every possible actual degree of p in (1), including leading-term cancellation.*

Proof. Lemma 3.1 gives $0 < \delta < 1$. Fix a witness and let $n = \deg x$, $m = \deg y \leq n$.

Case 1: $m < n$. Here $d = \deg p = n + 2$, because only $a_\delta x$ contributes its leading term. For $e > 0$, keep x, y fixed and set

$$\delta' = \delta + e, \quad p' = p - 2esx.$$

This gives the exact identity at δ' , and p' has degree d . Let M_x be the maximum absolute value of a coefficient of x ; it is positive. For example, any

$$0 < e < \min\{(1 - \delta)/2, \tau_p/(4M_x)\}$$

works by Lemma 6.1. The other two polynomials and the degree inequality are unchanged.

Case 2: $m = n$. No condition is imposed on $d = \deg p$. For $e > 0$, define

$$k = \sqrt{1 + e^2 + 2e/\delta}, \quad \delta' = \frac{k\delta}{1 + \delta e}, \quad \phi(s) = ks - e, \quad (16)$$

and the real mixing coefficients

$$\begin{aligned} \alpha &= 1 + e^2 + e(\delta + 1/\delta), & \beta &= e^2 + e/\delta, \\ \gamma &= e(1/\delta - \delta), & \theta &= 1 + e/\delta. \end{aligned} \quad (17)$$

They satisfy the exact identities

$$\begin{aligned} a_\delta(\phi(s)) &= \alpha a_{\delta'}(s) + \gamma b(s), \\ b(\phi(s)) &= \beta a_{\delta'}(s) + \theta b(s). \end{aligned} \quad (18)$$

For completeness the coefficient checks are

$$\begin{aligned} \alpha + \gamma &= k^2, & \alpha - \gamma &= 1 + e^2 + 2\delta e, & \alpha\delta' &= k(\delta + e), \\ \beta + \theta &= k^2, & \beta - \theta &= e^2 - 1, & \beta\delta' &= ke. \end{aligned}$$

Consequently the following are original polynomial witnesses if stable:

$$\begin{aligned} X(s) &= \alpha x(\phi(s)) + \beta y(\phi(s)), \\ Y(s) &= \gamma x(\phi(s)) + \theta y(\phi(s)), \\ P(s) &= p(\phi(s)), \quad P = a_{\delta'}X + bY. \end{aligned} \quad (19)$$

In particular, P is defined by composition with the actual polynomial p , not by treating its canceled coefficients as small nonzero ones.

As $e \downarrow 0$, all coefficients of X, Y, P tend respectively to those of x, y, p . For every e , $\deg X, \deg Y \leq n$ and $\deg P = d$, since $k > 0$. At $e = 0$ the leading coefficients of X, Y are the nonzero leading coefficients of x, y . Apply Lemma 6.1 at the respective degree bounds n, n, d . For all sufficiently small positive e this proves simultaneously

$$\deg X = \deg Y = n, \quad \deg P = d, \quad X, Y, P \text{ are nonzero and stable.}$$

More explicitly, take the three positive coefficient tolerances from that lemma and choose an integer j large enough that at $e = 2^{-j}$ the three finite coefficient differences are smaller than their respective tolerances. Coefficient continuity proves that every sufficiently large j works. This specifies a genuinely nonzero perturbation and does not presuppose any uniform stability margin.

Finally, directly from (16),

$$1 - (\delta')^2 = \frac{1 - \delta^2}{(1 + \delta e)^2}. \quad (20)$$

For every $e > 0$ the right side is strictly between 0 and $1 - \delta^2$. As $\delta' > 0$, this gives $\delta < \delta' < 1$. Stability is asserted only for the sufficiently small e just chosen, not for arbitrary positive e . The construction preserves real coefficients and the required degree inequality. No normalization is introduced. $\square$

6.3 Why cancellation is genuinely dangerous

At $\delta = 1/2$, the witness

$$x = s + \frac{3}{2}, \quad y = -s - \frac{1}{2}, \quad p = \frac{1}{2}s + 2 \quad (21)$$

is strictly stable and satisfies the original identity. Its nominal cubic expression has actual degree one. Changing only the parameter to $1/2 + e$ produces

$$p_e = -2es^2 + \left(\frac{1}{2} - 3e\right)s + 2.$$

For every $e > 0$ the product of its roots is $-1/e < 0$; thus it has a positive real root. Openness without an actual-degree constraint is not a valid argument.

The affine construction instead keeps $\deg P = 1$. For an exact stress test, take $e = 1/100$ and $k = \sqrt{10401}/100$. Formulas (16)–(19) give

$$\begin{aligned} \delta' &= \frac{\sqrt{10401}}{201} > \frac{1}{2}, \\ X &= \frac{201k}{200}s + \frac{30351}{20000}, \\ Y &= -\frac{201k}{200}s - \frac{9549}{20000}, \\ P &= \frac{k}{2}s + \frac{399}{200}. \end{aligned} \quad (22)$$

Each is visibly a nonzero stable linear polynomial, and the exact identity follows from (18). The inequality for δ' follows by squaring positive quantities: $4 \cdot 10401 > 201^2$. This test does not replace the all-degree proof.

No root at infinity is being ignored: the degrees of X, Y are secured by their nonzero leading coefficients, while the degree of $P = p \circ \phi$ is exactly the original actual degree. The original cancellation is therefore preserved in the identity for P , however many top coefficients have canceled.

7 Endpoint exclusion and the corrected feasible sets

Theorem 7.1 (Independent endpoint exclusion). *$P(Q)$ is false. More generally, $P(q)$ implies $q > Q$ for every $q \in I$.*

Proof. Suppose $P(q)$. By Theorem 6.2 there is a polynomial witness at some $\delta' > \delta(q)$ with $\delta' < 1$. Let $q' = (1 - \delta')/(1 + \delta')$. Then $0 < q' < q$ and $P(q')$. Lemma 3.2 gives $A(q')$, and the frozen equality $Q = \inf \mathcal{S}_A$ gives $Q \leq q' < q$. In particular $P(Q)$ is impossible. Only an infimum inequality is used after a strict polynomial perturbation has actually been constructed. $\square$

Proof of Theorem 2.1. There is no analytic feasible $q < Q$ by the definition of Q in (4); $A(Q)$ is a frozen input. For $q > Q$, scaling in Lemma 4.1 gives an analytic object at q . Hence $\mathcal{S}_A = [Q, 1)$. Equation (15) gives $P(q)$ for all $q > Q$, and Theorem 7.1 rules out every $q \leq Q$. Thus $\mathcal{S}_P = (Q, 1)$, including the stated rational realization.

If $P(q)$, its perturbed parameter $q' < q$ satisfies $A(q')$ by the preceding proof. Conversely, $A(r)$ at any $0 < r < q$ implies $P(q)$ by (15). This proves (8). The relative closure identity follows from the two explicit intervals. Finally Lemma 3.1 rules out all $\delta \geq 1$; on $(0, 1)$, the decreasing bijection (2) gives

$$q(\delta) > Q \iff 0 < \delta < \frac{1 - Q}{1 + Q}.$$

The bounds on Δ follow from $0 < Q < 1/2$. $\square$

7.1 The failed direct equivalence and its precise replacement

At $q = Q$, the frozen Core supplies

$$A(Q), \quad \forall N \ H_N(Q) \text{ feasible},$$

whereas Theorem 7.1 gives $\neg P(Q)$. Thus $P(q) \iff A(q)$ is false on all of I , even though $P(q) \implies A(q)$ always holds. It becomes a same-parameter equivalence on $I \setminus \{Q\}$. The exact all-parameter replacement, in the range of the hierarchy, is

$$P(q) \iff \exists r \in (0, q) \ \forall N \ H_N(r) \text{ feasible}, \quad 0 < q \leq 9/16. \quad (23)$$

This is a conclusion of the interface proof, not a change to H_N or to (3). The existential r can equivalently be required to be rational, since $P(q)$ implies $Q < q$ and a rational $r \in (Q, q)$ then exists.

Closure must be stated in the correct ambient space:

$$\overline{\mathcal{S}_P}^{(0,1)} = [Q, 1) = \mathcal{S}_A, \quad \overline{\mathcal{S}_P}^{\mathbb{R}} = [Q, 1].$$

The extra endpoint 1 in the second closure corresponds to $\delta = 0$, which is outside the original positive-parameter domain. In the original domain $(0, \infty)$, the pullback of analytic feasibility on $0 < \delta < 1$ is $(0, \Delta]$, and it is the relative closure of $\mathcal{A}_{\text{poly}}$.

7.2 Why the analytic endpoint creates no contradiction

An analytic function in $\mathcal{F}_{\sqrt{Q}}$ is required to have its specified fibres only on the open disc. It is not required to yield factors nonvanishing on a neighborhood of the closed disc. Indeed, such a stronger endpoint realization is impossible. If its factors u, v extended nonvanishingly to $|z| < R$ with $R > 1$, the identity (11) would continue there by the identity theorem. The function

$g(z) = Rz u(Rz)$ on $\mathbb{D}$ would then have its only simple zero at 0 and its only simple 1-points at $\pm i\sqrt{Q}/R$. It would establish $A(Q/R^2)$, contrary to $Q = \inf \mathcal{S}_A$.

In particular there is no closed-disc zero-free finite-factor certificate at Q . An arbitrary original polynomial witness is excluded separately by Theorem 7.1; it need not first be converted to a closed-disc object. No assertion that every endpoint function is transcendental, or any classification of boundary singularities, is needed.

8 Combination with the unchanged certified algorithm

The Core already specifies a program independently of the original admissibility predicate and proves that its nested strict intervals have the unique common point Q . Define Δ by (7), not by a decimal estimate. The present supplement supplies the missing logical connection to all original polynomial degrees and the endpoint:

$$\begin{array}{c}
 \text{frozen infinite-layer analytic criterion} \\
 \Downarrow \quad \text{frozen two-sided certificates and total precision program} \\
 \quad Q \text{ is exactly specified by that program} \\
 \Downarrow \quad \text{strict-slack realization and polynomial endpoint exclusion} \\
 \boxed{\delta \in \mathcal{A}_{\text{poly}} \iff 0 < \delta < \Delta, \quad \Delta = (1 - Q)/(1 + Q), \quad \Delta \notin \mathcal{A}_{\text{poly}}.}
 \end{array}$$

No same-parameter polynomial equivalence at Q occurs in this chain.

The same computed brackets give rational strict enclosures for Δ :

$$\frac{1 - R_j}{1 + R_j} < \Delta < \frac{1 - L_j}{1 + L_j}, \quad \frac{1 - L_j}{1 + L_j} - \frac{1 - R_j}{1 + R_j} = \frac{2(R_j - L_j)}{(1 + L_j)(1 + R_j)} \leq 2(R_j - L_j). \quad (24)$$

One extra bit of strict precision for Q therefore gives any requested strict precision for Δ . Total termination and the independent definition of Q are exactly the frozen ones, not new claims about a modified program.

Remark 8.1 (Meaning and limits of the combined algorithmic conclusion). The combination is not just an abstract assertion of computability: the fixed program uniquely specifies Q , the displayed transform uniquely specifies Δ , and both directions of the original feasibility criterion and the excluded endpoint have been proved. It makes no closed-form, algebraicity, complexity, or practical-runtime claim. Nor does it turn equality of arbitrary computable reals into a decidable question. With a promised nonendpoint effective real input, refining input and threshold intervals eventually decides the strict inequality. At a parameter specified to be Δ , infeasibility follows from the endpoint theorem, not from waiting for a negative finite layer. If Q were rational, the two certificate searches at that exact rational value would still need not terminate; this does not affect the frozen total precision algorithm. No stronger universal membership-decision claim is introduced here.

9 Adversarial self-audit and final verdict

The following checks concern the repair, not a renewed audit of the Core. Each item identifies the potential failure and the step preventing it.

1. *Reversed parameter direction.* Equation (20) proves $\delta' > \delta$, while $q'(\delta) = -2/(1 + \delta)^2 < 0$ for the map in (2). Thus a polynomial endpoint witness would cross to $q' < Q$, as required.
2. *A merely analytic perturbation.* Equations (18)–(19) are identities in $\mathbb{R}[s]$ and explicitly construct all three original polynomials. The perturbation is not a change only to an analytic representative.

3. *Leading-term cancellation or a root at infinity.* The exact example (21) defeats naive perturbation. Case 2 of Theorem 6.2 keeps the actual degree of p by composition, and fixes the degrees of X, Y by nonzero leading coefficients and Lemma 6.1. Arbitrarily many cancellations and constant p are included in that argument.
4. *Loss of strict stability under mixing.* Mixing is not claimed to preserve stability globally in e . The fixed-degree Rouché neighborhoods give simultaneous positive local margins; $e = 2^{-j} > 0$ eventually enters all three. Repeated roots need no genericity assumption.
5. *Loss of exact interpolation under rational approximation.* Only v is approximated, its constant term is fixed at $1/q$, and u_* is defined by exact polynomial division. The boundary estimate followed by the maximum modulus theorem controls the removable quotient at 0 as well as the whole disc.
6. *Real symmetry or degree constraints lost in realization.* Real Taylor coefficients and real rounding preserve symmetry. Equation (14) proves actual degrees, including artificial roots at $s = -1$ from denominator clearing. In particular both x and y have degree n , not merely degree at most n .
7. *An unavailable uniform margin as $q \downarrow Q$.* Every realization is made at a fixed strict pair $r < q$. Its disc radius $\sqrt{q/r} > 1$ and its positive minima may deteriorate at the endpoint. No argument exchanges this pointwise existence with a uniform estimate.
8. *Infimum used as endpoint exclusion.* Theorem 7.1 first constructs a strictly smaller feasible q' using original polynomials. Only then is the infimum inequality applied. Analytic attainment $A(Q)$ remains intact.
9. *Finite hierarchy secretly strengthened.* Equation (3) is unchanged at Q . Equation (23) inserts strict parameter slack as a derived polynomial-interface theorem and does not replace any frozen layer. No finite stabilization is used.
10. *Finite computation used in place of proof.* The Cayley identities, mixing identities, direction identity, and example (22) were also checked by exact symbolic arithmetic (SymPy 1.14.0, rational expressions with $k^2 = 1 + e^2 + 2e/\delta$). The coefficient identities and analytic arguments above are the proofs; neither floating-point roots nor these checks serve as a substitute.

PASS — endpoint/interface repaired.

This verdict is for the endpoint/interface repair with the Core results accepted as frozen inputs. It does not assert that those proofs were reaudited in this round. The combined mathematical statement is exactly Theorem 2.1: the analytic parameter set is $[Q, 1)$, the polynomial parameter set is $(Q, 1)$, and the original admissible set is $(0, (1 - Q)/(1 + Q))$. The finite polynomial construction and the independent endpoint exclusion close the interface without changing the Core.

A Frozen-source identification

The original problem used here is the algebraic definition in

`D:/Downloads/belgian_chocolate_problem_description.txt`

The accepted Core source is

`D:/ai4sc/belgian_chocolate_complete_algorithmic_solution.md`

Its SHA-256 fingerprint at the start of this supplement was

B846BD95A1DB94C471BC313C319E8E535
F29BE8BC23C1B7F2F41BA3F02B296D0

The line references below identify frozen inputs only; they do not import an unproved polynomial-interface equivalence.

- Section 1, lines 31–41: $\mathcal{F}_t$, the analytic parameter set and its infimum.
- Theorem 3, lines 244–247: the all-layer criterion on $0 < q \leq 9/16$.
- Section 5, formula (T), lines 314–320: analytic endpoint attainment and $0 < q_* < 1/2$.
- Corollary 4 and Theorem 5, lines 322–324 and 348: the two finite-certificate sides.
- Theorem 6 and Section 9, lines 415–439: total subdivision, effective error bound, independent definition of Q , and $Q = q_*$.
- Section 11: non-stabilization of any fixed finite layer.

The prior referee report and the Core source are not edited by this supplement. Statements in this supplement are proved from the original definition and the explicitly listed accepted inputs; earlier agent judgments are not proof premises.